CORK INSTITUTE OF TECHNOLOGY
INSTITIÚID TEICNEOLAÍOCHTA CHORCAÍ

Semester Examinations 20XX/XX

Module Title: Times Series & PCA

Module Code: STAT9005

School: Mathematical Science and Informatics

Programme Title: MSc in Science in Data Science and Analytics

Internal Examiner(s): Francisco Hernandez

Instructions: Answer **ALL** questions. **Section A**. MCQ / True or False questions (50%) **Section B.** Practical part (50%). In Section A, questions have a **–0.5** negative mark for an incorrect answer, **0** for a question not attempted and a **+0.5** positive mark for a correct answer.

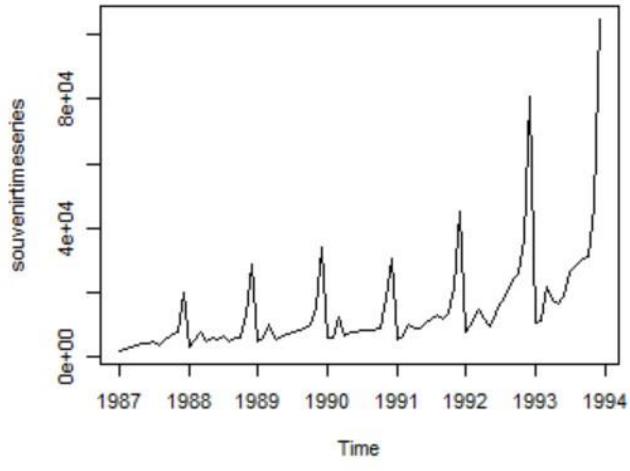
Duration: XX minutes

Section A: MCQ and True or false questions (10 marks)

1. The fluctuations that occurs in a time series can be classified into 4 basic types of variation: Secular trend, Seasonal variation, Cyclical variation and Residual variation. While the first three (trend, seasonal and cyclical variation) are totally deterministic, the residual variation is known as the random component of the series.

True: _____ or False: _____.
If false, state the correct answer: _____
_____

2. According to the next figure of a time series distribution. We can state that the series shows a seasonal pattern repeated every year following a typical additive model of the form **A = T + S + C + R.**



True: _____ or False: _____.
If false, state the correct answer: _____
_____

3. The sales figures (in thousands of €) of a retailing company for the years 2009, 2010 and 2011 are shown in the next table.

YEAR	QUARTER			
	I	II	III	IV
2009	658	601	547	672
2010	648	618	524	656
2011	550	528	415	534

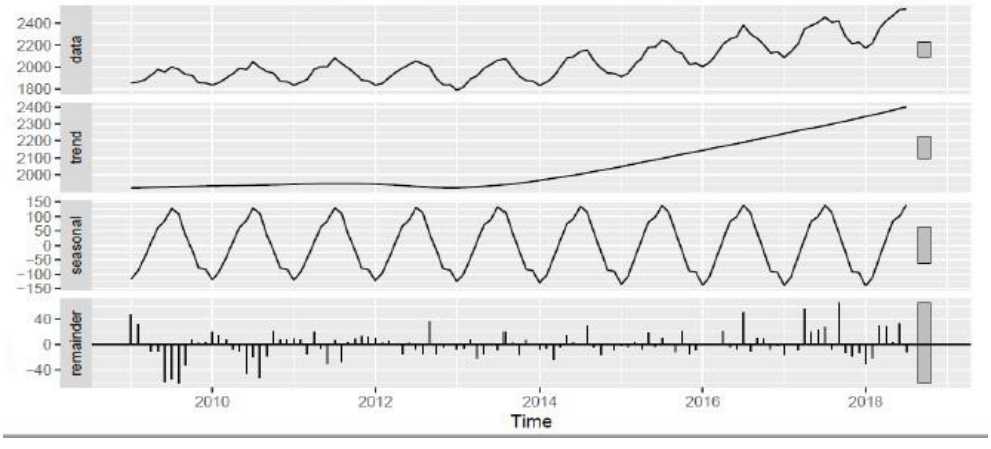
Identify the value of "p" indicating the periodicity of the time series.

A: p = 3 B: p = 4 C: p = 5 D: The series has no periodicity.

4. Double exponential smoothing methods – Brown's, Holt's and Damped methods – are an extension of the simpler smoothing methods to model the trend component of the time series.

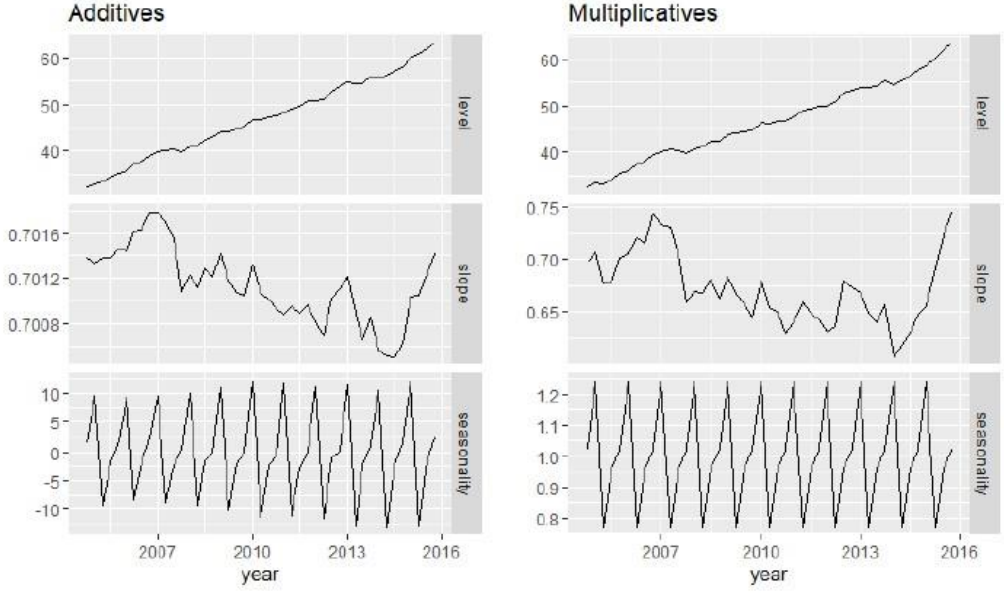
True: _____ or False: _____.
If false, state the correct answer: _____

5. According to the following decomposed time series, we can conclude that the seasonal component has a higher effect on the variability of the series than the trend component.



True: _____ or False: _____.
If false, state the correct answer: _____

6. After the applying a triple exponential smoothing test to a dataset, the following figure shows the decomposition of the parameters under the assumption of the time series following an "Additive" or a "Multiplicative" model.



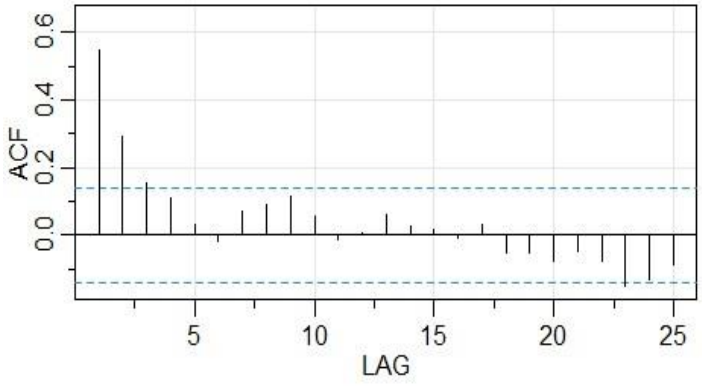
Based on the figure, which model describes better the time series? Justify your answer.

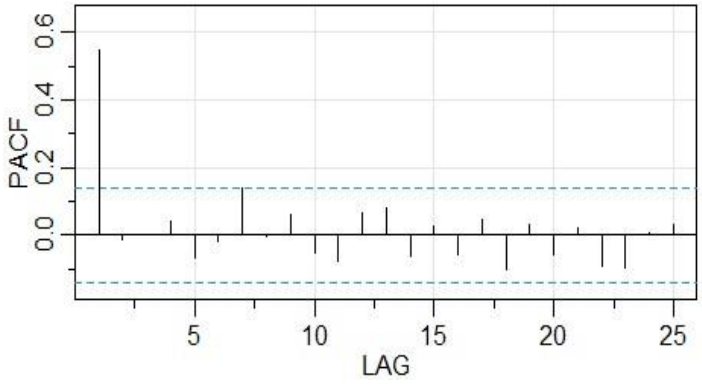
A. Additive B. Multiplicative

_____

_____

7. According to the following ACF and PACF of a time series, identify what type of ARIMA(p,d,q) process identify that series.





A. MA(1) B. ARIMA(2,1,0) C. AR(1) D. AR(2)

Section B: Practical part (10 marks)

1. **QUESTION 1** **(1 mark)**

   Simulate a *AR(2)* using the command *"arima.sim"*. Plot the process and the correlogram. Do the correlograms show what you expected for the value of the parameter chosen for this *ARIMA(p,0,0)* process?

2. **QUESTION 2** **(9 marks)**

   Analysis of the Time Series *"AirPassengers"* in *RStudio*. Load the libraries "*astsa*", "*forecast*" and "*tseries*" for the analysis.

   a. Plot the time series and comment on the graph. **(0.75 marks)**

   b. Use the "*decompose*" function to analyse the components of the series (chose appropriate model: additive or multiplicative). Comment on the results of the graph. Which model do you think is better to describe this series? Which of the components has a bigger impact on the time series *AirPassengers*? Explain your conclusions.
   (1.5 marks)

   c. Use the "*mstl*" function to find the periodicity of the time series, if any. Plot the results.
   (0.75 marks)

   # Classical Methods

   d. According to your previous analysis, which of the Classical Method (*SES*, *Holt's*, *Holt-Winter's*) do you consider is more appropriate for describing this data.
   (0.75 marks)

   e. Apply the Classical Method you chose in the previous section using both model options, additive and multiplicative (*R* commands "*ses*", "*holt*" or "*hw*").
   (1.5 marks)

   f. Which model do you think will better explain this time series, additive or multiplicative? Use GOF measures to support your observations. (use R command "*accuracy*")
   (0.75 marks)

   # ARIMA models

   g. Is the time series stationary? What do you expect from a stationary time series?
   (0.75 marks)

   h. Remove the variability of the series by applying logs.
   Remove the trend of the log- transformed series by using "*diff*" command.
   Plot transformed time series using "*decompose*". Do you find any issues with seasonality?
   What is the lag interval for this periodicity?
   You can use the help of ACF/PACF (use "*acf2*").
   (1.5 marks)

   i. Use "*auto.arima()*" function to find the best possible *ARIMA(p,d,q) x SARIMA(P,D,Q)s*. Is this result in line with what you expected from your previous analysis?
   (0.75 marks)